ABBA2 Implicit: Simultaneous State–Multiplier Formulation

Simultaneous Newton formulation for the output state and multiplier

GC2D implementation of equation (21) from the original implicit note

1 Purpose and relation to the reduced formulation

The `simultaneous_state_multiplier` option of `ABBA2Implicit` computes the same exact projected map as its `reduced_multiplier` option but uses a different nonlinear system. The reduced formulation eliminates the duplicated output state and solves only for the two-component multiplier. The simultaneous formulation retains both the four-component duplicated output and the multiplier as Newton unknowns. Consequently, for one guiding-centre particle, each Newton correction solves a 6×6 system rather than the reduced 2×2 Schur complement.

The two formulations must converge to the same physical state whenever the projection root is locally unique and both nonlinear solves converge to that root. Finite tolerances and floating-point linear algebra can produce differences at the round-off level.

2 Guiding-centre and duplicated states

The physical state $z = (X, Y) \in \mathbb{R}^2$ satisfies

$$\dot{z} = f(z,t), \qquad f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Its spatial Jacobian

$$W(z,t) = D_z f(z,t) = J_0 \nabla_z^2 \phi(z,t)$$

satisfies $W^T J_0 + J_0 W = 0$ because the Hessian of ϕ is symmetric.

The duplicated state and physical diagonal are

$$\mathcal{Y} = (u,v) \in \mathbb{R}^4, \qquad \mathcal{M} = \{(u,v) : u = v\}.$$

Define

$$G = (I_2 \ \ -I_2), \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}, \qquad E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}.$$

Then $Ez = (z,z)$ is diagonal, $N\mu = (\mu, -\mu)$ is a correction normal to the diagonal, and

$$GE = 0, \qquad GN = GG^T = 2I_2.$$

3 Endpoint-time ABBA map

Let $s = h/2$ and $t_{n+1} = t_n + h$. The nonautonomous ABBA map is

$$\Phi_{h,t_n}^{\text{ABBA}} = A_{s,t_{n+1}} \circ B_{s,t_{n+1}} \circ B_{s,t_n} \circ A_{s,t_n},$$

where

$$A_{s,t}(u,v) = (u + sf(v,t), v), \qquad B_{s,t}(u,v) = (u, v + sf(u,t)).$$

Starting from $(u^{(0)}, v^{(0)})$, the four explicit stages are

$$\begin{aligned}
u^{(1)} &= u^{(0)} + sf(v^{(0)}, t_n),\\
v^{(1)} &= v^{(0)} + sf(u^{(1)}, t_n),\\
v^{(2)} &= v^{(1)} + sf(u^{(1)}, t_{n+1}),\\
u^{(2)} &= u^{(1)} + sf(v^{(2)}, t_{n+1}).
\end{aligned}$$

Thus $\Phi^{\mathrm{ABBA}}_{h,t_n}(u^{(0)}, v^{(0)}) = (u^{(2)}, v^{(2)})$. The endpoint-time order makes the map symmetric:

$$\Phi^{\mathrm{ABBA}}_{-h,t_{n+1}} = \left(\Phi^{\mathrm{ABBA}}_{h,t_n}\right)^{-1}.$$

At the stage points, define

$$W_1 = W(v^{(0)}, t_n), \quad W_2 = W(u^{(1)}, t_n), \quad W_3 = W(u^{(1)}, t_{n+1}), \quad W_4 = W(v^{(2)}, t_{n+1}),$$

and $S = W_2 + W_3$. The exact duplicated ABBA Jacobian is

$$D\Phi^{\mathrm{ABBA}}_{h,t_n} = \begin{pmatrix} I_2 + s^2 W_4 S & s(W_1 + W_4) + s^3 W_4 S W_1 \\ sS & I_2 + s^2 S W_1 \end{pmatrix}.$$

The matrix products retain this order because stage Jacobians evaluated at different points need not commute.

4 Simultaneous projection equations

Given the diagonal input $\mathcal{Y}_n = Ez_n$, pre-correct the two copies by the unknown multiplier:

$$\widehat{\mathcal{Y}}_n(\mu) = \mathcal{Y}_n + N\mu = (z_n + \mu, z_n - \mu).$$

The simultaneous formulation retains the diagonal output $\mathcal{Y}_{n+1}$ as an additional unknown and imposes

$$\begin{aligned}
d(\mathcal{Y}_{n+1}, \mu) &:= \mathcal{Y}_{n+1} - N\mu - \Phi^{\mathrm{ABBA}}_{h,t_n}(\mathcal{Y}_n + N\mu) = 0,\\
g(\mathcal{Y}_{n+1}) &:= G\mathcal{Y}_{n+1} = 0.
\end{aligned}$$

The first equation states that the same multiplier used to separate the input copies is applied with the opposite placement around the ABBA output. The second equation constrains the final copies to the physical diagonal.

Writing $\mathcal{Y}_{n+1} = (y_u, y_v)$, the residual components used by the implementation are

$$d_u = y_u - \mu - u^{(2)}(\mu), \qquad d_v = y_v + \mu - v^{(2)}(\mu), \qquad g = y_u - y_v.$$

5 Exact simultaneous Newton system

Because G and $N = G^T$ are constant,

$$D_{\mathcal{Y}_{n+1}} d = I_4, \qquad D_\mu d = -\left[I_4 + D\Phi^{\mathrm{ABBA}}_{h,t_n}(\widehat{\mathcal{Y}}_n)\right] N,$$

$$D_{\mathcal{Y}_{n+1}} g = G, \qquad D_\mu g = 0.$$

Therefore, at Newton iterate $(\mathcal{Y}^{(k)}_{n+1}, \mu^{(k)})$, the simultaneous formulation solves

$$\boxed{\begin{pmatrix} I_4 & -\left[I_4 + D\Phi^{\mathrm{ABBA}}_{h,t_n}(\widehat{\mathcal{Y}}_n)\right] G^T \\ G & 0 \end{pmatrix} \begin{pmatrix} \Delta\mathcal{Y}_{n+1} \\ \Delta\mu \end{pmatrix} = -\begin{pmatrix} d(\mathcal{Y}_{n+1}, \mu) \\ g(\mathcal{Y}_{n+1}) \end{pmatrix}.} \tag{21}$$

The number (21) is retained deliberately to identify the equation from the original `ABBA_implicit` document.

The update is

$$\mathcal{Y}_{n+1}^{(k+1)} = \mathcal{Y}_{n+1}^{(k)} + \Delta\mathcal{Y}_{n+1}, \qquad \mu^{(k+1)} = \mu^{(k)} + \Delta\mu.$$

No inverse is formed explicitly; a dense linear solve is used for each 6×6 particle block.

6 Schur complement and equivalence to the reduced formulation

Eliminating $\Delta\mathcal{Y}_{n+1}$ from (21) gives the Schur complement

$$\mathcal{K} = G\left[I_4 + D\Phi_{h,t_n}^{\mathrm{ABBA}}(\widehat{\mathcal{Y}}_n)\right]G^T = 2I_2 + GD\Phi_{h,t_n}^{\mathrm{ABBA}}(\widehat{\mathcal{Y}}_n)G^T.$$

Let the reduced residual be

$$R(\mu) = G\Phi_{h,t_n}^{\mathrm{ABBA}}(\mathcal{Y}_n + G^T\mu) + 2\mu.$$

Then

$$\mathcal{K} = R'(\mu).$$

Moreover, from the definitions of d and g,

$$Gd = g - R(\mu).$$

The second block row of (21), after substituting the first one, therefore yields

$$\mathcal{K}\,\Delta\mu = -R(\mu).$$

This is exactly the Newton correction used by the reduced formulation. Hence both algorithms generate the same multiplier iterates in exact arithmetic when they start from the same $\mu^{(0)}$, even though the simultaneous formulation also updates the duplicated output unknown.

7 Initial iterate and convergence criterion

The implementation uses

$$\mu^{(0)} = 0, \qquad \mathcal{Y}_{n+1}^{(0)} = \Phi_{h,t_n}^{\mathrm{ABBA}}(\mathcal{Y}_n).$$

This makes $d = 0$ initially while leaving the diagonal defect g to drive the first Newton correction. For a physical input scale

$$s_z = \max(1, \|z_n\|_\infty),$$

the iteration stops when

$$\max\left(\|d\|_\infty, \|g\|_\infty\right) \le \mathrm{atol} + \mathrm{rtol}\, s_z.$$

Newton is not damped, so the residual norm is not required to decrease monotonically. Failure to converge within the configured iteration count, or a singular simultaneous Jacobian, terminates the integration step with an error.

For N_p independent guiding-centre particles, physical states use component-major storage,

$$(X_1, \ldots, X_{N_p}, Y_1, \ldots, Y_{N_p}),$$

but equation (21) is assembled as N_p independent 6×6 systems. This preserves the absence of artificial cross-particle derivatives while allowing a vectorized solve.

At convergence, y_u and y_v agree up to the stopping tolerance. The physical state is represented neutrally by

$$z_{n+1} = \frac{y_u + y_v}{2}.$$

8 Symmetry and symplecticity of the converged map

At the exact root, both formulations satisfy

$$\mathcal{Y}_{n+1} - N\mu = \Phi_{h,t_n}^{\mathrm{ABBA}}(\mathcal{Y}_n + N\mu), \qquad \mathcal{Y}_n, \mathcal{Y}_{n+1} \in \mathcal{M}.$$

Using the inverse ABBA map and defining the reverse multiplier as $\mu_{\mathrm{inv}} = -\mu$ gives

$$\Psi_{-h,t_{n+1}} \circ \Psi_{h,t_n} = I,$$

provided the root is locally unique and solved exactly.

The duplicated symplectic form is

$$\Omega_\times = \begin{pmatrix} 0 & J_0 \\ J_0 & 0 \end{pmatrix}.$$

Every A and B shear preserves $\Omega_\times$, and therefore so does $\Phi_{h,t_n}^{\mathrm{ABBA}}$. The diagonal and correction directions satisfy

$$E^T\Omega_\times E = 2J_0, \qquad N^T\Omega_\times N = -2J_0, \qquad E^T\Omega_\times N = N^T\Omega_\times E = 0.$$

Differentiating the exact projected equation with $M = D\mu_h(z)$ and $J_\Psi = D\Psi_{h,t_n}(z)$ gives

$$D\Phi_{h,t_n}^{\mathrm{ABBA}}(E + NM) = EJ_\Psi - NM.$$

Preservation of $\Omega_\times$ and the orthogonality identities cancel the terms containing M, leaving

$$\boxed{J_\Psi^T J_0 J_\Psi = J_0.}$$

Thus the simultaneous formulation is the same symmetric and symplectic physical map as the reduced formulation at the exact root. In finite arithmetic, measured reversibility and symplecticity also include the simultaneous Newton tolerance and linear-solve round-off.

9 Implementation correspondence

The production implementation follows the mathematical blocks directly:

- the endpoint-time stages evaluate $\Phi_{h,t_n}^{\mathrm{ABBA}}$;
- their exact derivatives assemble $D\Phi_{h,t_n}^{\mathrm{ABBA}}$;
- each particle receives the 6×6 matrix in equation (21);
- diagnostics record the simultaneous residual norm, iteration count, and multiplier norm;
- the optional ideal physical tangent is evaluated by implicit differentiation of the common exact root, independently of the finite Newton stopping decision.